

Habituation-Aware Spatio-Temporal Logic for Predictive Task Allocation in Decentralized Multi-Robot Vineyard Bird Deterrence

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MS Thesis Proposal

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Abstract

Mobile multi-robot teams are an attractive alternative to static scare devices for protecting vineyards from crop-damaging birds, because static and fixed-schedule deterrents lose effectiveness as birds *habituate*. A working decentralized simulation testbed already exists for this problem: it couples an online self-exciting spatio-temporal point-process (SESTPP) risk model, health-weighted zone partitioning, and a reactive/predictive task pipeline. However, across extensive 24-hour sweeps the predictive branch of that pipeline does not outperform a simple greedy unconstrained dispatcher on the primary mission metrics. The diagnosis is not that predictive capacity is starved; it is that each predictive task does not carry enough *operational value* under a scalar value-weighted-exposure objective. This proposal attacks that gap directly. We argue that the true mission objective is not a scalar but a *spatio-temporal specification*, and that the single most important missing term is habituation. We (i) introduce an explicit per-location, per-cue habituation model into the ground-truth process and the planner’s forward predictor, and (ii) re-define predictive task value as the *counterfactual robustness margin* of a Signal Temporal Logic (STL) mission specification that combines bounded exposure, reactive deadlines, coverage recency, and an anti-habituation (cue-variety) requirement. The two contributions are complementary: habituation makes the objective irreducible to greedy exposure, and STL robustness is the mechanism that turns that objective into a usable, gradient-bearing task score. The work reuses the existing decentralized architecture; the predictive value function is the only component replaced. We give the full formulation, the monitoring and scoring algorithms, and a falsifiable experimental design with a baseline ladder isolating each contribution.

1 Introduction and Motivation

Frugivorous and granivorous birds inflict substantial losses on wine and table-grape growers. The standard countermeasures (netting, sound cannons, distress-call broadcasts, visual repellents, and fixed cue-rotation schedules) are limited in characteristic ways. Netting is reliable but carries large seasonal deployment and removal costs; acoustic and visual deterrents work only transiently because birds *habituate* to repeated, predictable stimuli and resume foraging [14]. Fixed or manually tuned cue schedules do not adapt to where and when birds are active, nor to the recent history of deterrence at a location. A mobile multi-robot team that senses bird activity and dispatches deterrence actions adaptively is therefore a natural and increasingly feasible alternative.

A decentralized simulation testbed for this problem already exists and is the starting point of this thesis. It models a row-structured vineyard with a value-weighted spatial surface, a heterogeneous UAV/UGV fleet with capability, endurance, and health attributes, health-weighted zone partitioning with a neighbor graph, an online per-robot SESTPP risk field, and a task pipeline that generates,

selects, assigns, and dispatches two streams of tasks: *reactive* tasks triggered by confirmed detections and *predictive* (preventive) tasks generated from forecast hotspots. The original framing advertised the intervention-aware SESTPP itself as the headline contribution.

Two problems motivate a redefinition of the thesis. First, the SESTPP model is not novel: it is a spatio-temporal Hawkes process [12, 13] augmented with an inhibitory term, and self-exciting models with suppression are well established. Second, and more importantly, the empirical results are unsatisfactory. Across nominal 24-hour sweeps, every variant of the predictive pipeline (lead-time reservation, adaptive gating, time-aware selection, and a centralized-global topology) fails to beat a greedy unconstrained dispatcher on the primary metrics (value-weighted exposure, reactive response quality, travel efficiency). The recurring finding is summarized cleanly in the project’s own analysis: the bottleneck is *not* preserving predictive share but *increasing the operational value of predictive work*.

We take that diagnosis literally. Under a scalar value-weighted-exposure objective, a greedy policy that always serves the nearest, highest-value detection is hard to beat, because exposure reduction is local, immediate, and myopic-friendly. Prediction can only pay when the objective rewards something a myopic reactive policy structurally cannot optimize. Habituation is exactly such a quantity: if a deterrent loses effectiveness when the same cue is applied repeatedly at the same place, then the value of an action depends on the *history* of actions at that location and on the *variety* of cues used (a temporal, non-local property). A policy that anticipates and spreads deterrence in space, time, and cue-type maintains effectiveness; a greedy policy that hammers the nearest hotspot with whatever cue is available degrades it.

This observation suggests two coupled changes. The objective should be expressed as a *spatio-temporal specification* rather than a scalar, and the dominant missing clause in that specification is anti-habituation. We adopt Signal Temporal Logic (STL) [1, 2, 3] because its quantitative (robustness) semantics yields a real-valued satisfaction margin that can serve directly as a task value, and because it expresses timed, history-dependent requirements naturally.

Contributions. This thesis proposes:

1. An explicit *habituation model* over (location, cue-mode) pairs, inserted both into the ground-truth bird process (so habituation is real in the world) and into the planner’s short-horizon forward predictor (so the planner can reason about it).
2. A *habituation-aware STL mission specification* Φ combining bounded value-weighted exposure, reactive deadline satisfaction, coverage recency, and an anti-habituation cue-variety clause, with a smooth conjunctive robustness so the score is gradient-bearing and does not collapse to the flat objective that defeated the prior hand-tuned value function.
3. A *decentralized predictive task-value function* defined as the counterfactual robustness margin a candidate action contributes to the local sub-specification, computed from the existing SESTPP field plus the habituation state, and dropped into the existing reserved-capacity dispatcher in place of the prior ΔJ score.
4. A *falsifiable evaluation*: a baseline ladder isolating the robustness reformulation from the habituation objective, run under both habituating and non-habituating ground truth, so the source of any advantage is identifiable.

The SESTPP model, zone partitioner, fleet model, and dispatcher are retained as supporting infrastructure. The predictive value function is the only component this thesis replaces.

2 Background and Related Work

Multi-robot task allocation (MRTA). The reactive/predictive setting here is a single-task-robot, single-robot-task, time-extended assignment problem with two task classes of differing temporal semantics, placing it in the ST-SR-TA region of the standard MRTA taxonomy [9]. The existing dispatcher already implements a reserved-capacity policy with a single reservation parameter and a hard-latency reactive override; we reuse that allocation layer and modify only the value attached to predictive tasks.

Temporal logic for robotics. Linear Temporal Logic (LTL) and its metric and signal extensions are widely used to specify and verify robot missions [11, 10]. STL [1] interprets formulas over real-valued, continuous-time signals and admits a *quantitative* semantics: a robustness degree measuring how strongly a trajectory satisfies (positive) or violates (negative) a formula [2, 3]. Because min/max-based robustness has zero gradient almost everywhere, smooth surrogates (softmin/softmax and the arithmetic-geometric-mean (AGM) robustness) have been introduced for optimization and control [4, 5, 6]. Online and past-time monitoring of temporal specifications over cyber-physical systems is surveyed in [7]; spatial extensions that reason over networks of locations, such as SSTL and the spatio-temporal reach/escape family, target exactly distributed sensor/robot systems [8]. This thesis uses bounded-future STL for operational clauses and past-time STL for the history-dependent anti-habitation clause, and adopts smooth robustness for scoring.

Self-exciting point processes. The SESTPP risk model is a discrete-time spatio-temporal Hawkes process with an inhibitory term [12, 13]. In this proposal it is demoted from headline contribution to a *principled predictive workload generator and forward predictor*; it supplies the intensity field used inside the counterfactual robustness computation, but it is not itself the claimed novelty.

Habituation. Habituation to repeated acoustic and visual deterrents is a long-documented phenomenon in wildlife management [14]; effectiveness typically declines with repeated, predictable exposure and partially recovers after a stimulus-free interval, and varying or rotating cues delays decline. We capture this with a stylized per-(location, cue) effectiveness state. We do not attempt to model bird cognition; the model only needs the qualitative property that repetition degrades and absence restores effectiveness, which is sufficient to make variety and anticipation valuable.

3 System Model

Domain and fleet. The vineyard is a bounded domain $\Omega \subset \mathbb{R}^2$ over a horizon $[0, T]$. A heterogeneous fleet $\mathcal{R} = \{r_1, \dots, r_N\}$ operates over Ω ; robot r has position $p_r(t) \in \Omega$, translational speed v_r , a busy flag $b_r(t) \in \{0, 1\}$, and a local task queue $Q_r(t)$. Robots are partitioned into zones $\{Z_r\}_{r \in \mathcal{R}}$ by the existing health-weighted power-diagram partitioner, with a neighbor graph $G = (\mathcal{R}, E)$; ∂Z_r denotes the boundary band of width d_b over which robot r exchanges events with neighbors.

Cue modes. Deterrence actions are drawn from a finite set of cue modes $\mathcal{M} = \{1, \dots, m\}$ (e.g. acoustic burst, distress call, UAV flyby, visual strobe). A deterrence action is a tuple $a = (x_a, c_a, r)$: location $x_a \in \Omega$, mode $c_a \in \mathcal{M}$, executing robot r .

Value surface. Each point carries a static value weight (existing model)

$$w(x) = 1 + w_{\text{row}}(x) + w_{\text{edge}}(x), \quad (1)$$

where w_{row} rewards proximity to vineyard-row centerlines and $w_{\text{edge}}(x) = g_{\text{edge}} \exp(-d_{\text{edge}}(x)/\ell_{\text{edge}})$ rewards proximity to the field boundary. The same surface is reused in scoring and in the exposure metric.

Two task streams. Following the existing pipeline, two streams arrive over time:

- *Reactive* tasks $\mathcal{A}_{\text{react}}$ are triggered by confirmed detections. Each $a \in \mathcal{A}_{\text{react}}$ has a value $v_a \geq v_{\min}$, an arrival time t_a^{arr} , a hard deadline $t_a^{\text{exp}} = t_a^{\text{arr}} + T_{\text{react}}$ (after which it expires and contributes value-weighted exposure), and a target x_a .
- *Predictive* tasks $\mathcal{A}_{\text{pred}}$ are generated asynchronously from the SESTPP field. Each carries a candidate action (x_a, c_a) , a service-cost surrogate $c_\eta(a, r)$, and no hard deadline. Predictive tasks may be dropped without direct exposure, but foregoing a good one forgoes future exposure reduction.

The service time of action a on robot r is

$$\tau_{\text{service}}(a, r) = \frac{\|p_r - x_a\|}{v_r} + t_{\text{spin}}(a, r) + t_{\text{action}}(a), \quad (2)$$

with type-dependent spin-up and action durations. The scenario-characterizing quantity is the reactive load factor

$$\rho_{\text{load}} = \frac{\lambda_{\text{react}} \mathbb{E}[\tau_{\text{service}, \mathcal{R}}]}{N}, \quad (3)$$

where λ_{react} is the mean reactive arrival rate. Values $\rho_{\text{load}} < 1$ indicate spare capacity; $\rho_{\text{load}} \approx 1$ indicates reactive-bound saturation; $\rho_{\text{load}} > 1$ means predictive work necessarily displaces reactive service. The proposed policy is expected to matter near $\rho_{\text{load}} \approx 1$.

4 Habituation Model

We introduce a per-(zone, cue-mode) effectiveness state. Let Z index a spatial cell (reusing the planner grid) and $c \in \mathcal{M}$ a cue mode. The effectiveness $\eta_{Z,c}(t) \in (0, 1]$ scales the suppressive power of a mode- c action applied in cell Z at time t .

Recovery and habituation dynamics. Between applications, effectiveness recovers exponentially toward 1 with time constant T_{rec} :

$$\frac{d\eta_{Z,c}}{dt} = \frac{1 - \eta_{Z,c}(t)}{T_{\text{rec}}} \iff \eta_{Z,c}(t + \Delta t) = 1 - (1 - \eta_{Z,c}(t)) e^{-\Delta t/T_{\text{rec}}}. \quad (4)$$

On a completed mode- c action in cell Z at time τ , effectiveness drops multiplicatively by a per-mode habituation increment $\kappa_c \in (0, 1)$:

$$\eta_{Z,c}(\tau^+) = \eta_{Z,c}(\tau^-) (1 - \kappa_c). \quad (5)$$

Optionally, cross-mode generalization couples modes: a mode- c application also habituates $c' \neq c$ by $\gamma \kappa_c$ with a small coupling $\gamma \in [0, 1)$, $\eta_{Z,c'}(\tau^+) = \eta_{Z,c'}(\tau^-) (1 - \gamma \kappa_c)$. Setting $\gamma = 0$ recovers independent modes; $\gamma > 0$ makes pure rotation between two modes insufficient and rewards genuine spatial and temporal spreading.

Effective suppression in the ground-truth process. The existing ground-truth bird intensity is a discrete-time self-exciting field with suppression from completed deterrence. We make that suppression *effectiveness-scaled*. With background $\mu(x)s(t)$, offspring kernel K_σ , and suppression kernel K_{σ_u} ,

$$\lambda_{\text{true}}(x, t) = \left[\mu(x) s(t) + \sum_{(x_i, t_i) \in \mathcal{E}} \alpha K_\sigma(x - x_i) e^{-(t-t_i)/\omega} - \sum_{(z_j, \tau_j, c_j) \in \mathcal{U}} \underbrace{\eta_{z_j, c_j}(\tau_j)}_{\text{habituation}} \beta_{c_j} K_{\sigma_u}(x - z_j) e^{-(t-\tau_j)/\omega_u} \right]_+, \quad (6)$$

where $\mathcal{E}$ is the detection/event history, $\mathcal{U}$ the completed-intervention history, β_c the nominal mode- c suppression strength, and $[\cdot]_+$ the positive part. Equation (6) is the crux of the redesign: a deterrence applied with low current effectiveness $\eta_{z,c}$ suppresses little, so a policy that repeatedly applies the same cue at the same place lets exposure rebuild, while a policy that maintains high effectiveness (through variety and spacing) suppresses more for the same number of actions.

Primary outcome. The value-weighted exposure remains the headline outcome metric:

$$J_{\text{exp}} = \int_0^T \int_{\Omega} w(x) \lambda_{\text{true}}(x, t) dx dt. \quad (7)$$

Crucially, under (6) a habituation-blind policy and a habituation-aware policy produce *different* J_{exp} even when they perform the same number of actions, which is what makes predictive variety operationally valuable.

5 Habituation-Aware Spatio-Temporal Specification

5.1 Signals

We monitor a small set of real-valued signals derived from system state, sampled at a monitor period Δ_m . For a cell or zone Z :

$$e_Z(t) = \int_Z w(x) \hat{\lambda}_r(x, t) dx \quad (\text{value-weighted exposure rate, from SESTPP estimate } \hat{\lambda}_r) \quad (8)$$

$$\text{slack}_a(t) = t_a^{\text{exp}} - t \quad (\text{time-to-deadline of open reactive task } a) \quad (9)$$

$$g_Z(t) = t - t_Z^{\text{visit}} \quad (\text{time since cell } Z \text{ last serviced}) \quad (10)$$

$$\eta_{Z,c}(t) \quad (\text{per-mode effectiveness from Sec. 4})$$

5.2 STL syntax

We use STL with bounded-future and past-time temporal operators. Atomic predicates have the form $\mu \equiv (f(\xi(t)) \geq 0)$ for a signal map f . The grammar is

$$\varphi ::= \top \mid \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \mathcal{U}_I \varphi_2 \mid \varphi_1 \mathcal{S}_I \varphi_2, \quad (11)$$

with bounded interval $I \subset \mathbb{R}_{\geq 0}$. Derived operators: future eventually $\Diamond_I \varphi \equiv \top \mathcal{U}_I \varphi$, future always $\Box_I \varphi \equiv \neg \Diamond_I \neg \varphi$; and past-time *once* $\Diamond_I \varphi$ and *historically* $\Box_I \varphi$ defined dually from the since operator $\mathcal{S}_I$. Past-time operators are used for the history-dependent anti-habituation clause and are convenient for online monitoring because they depend only on the observed prefix.

5.3 Quantitative (robustness) semantics

The robustness $\rho(\varphi, \xi, t) \in \mathbb{R}$ of formula φ over signal ξ at time t is defined recursively [2, 3]:

$$\rho(\mu, \xi, t) = f(\xi(t)) \quad (12)$$

$$\rho(\neg\varphi, \xi, t) = -\rho(\varphi, \xi, t) \quad (13)$$

$$\rho(\varphi_1 \wedge \varphi_2, \xi, t) = \min(\rho(\varphi_1, \xi, t), \rho(\varphi_2, \xi, t)) \quad (14)$$

$$\rho(\Box_I \varphi, \xi, t) = \inf_{t' \in t \oplus I} \rho(\varphi, \xi, t') \quad (15)$$

$$\rho(\Diamond_I \varphi, \xi, t) = \sup_{t' \in t \oplus I} \rho(\varphi, \xi, t') \quad (16)$$

$$\rho(\varphi_1 \mathcal{U}_I \varphi_2, \xi, t) = \sup_{t' \in t \oplus I} \min\left(\rho(\varphi_2, \xi, t'), \inf_{t'' \in [t, t']} \rho(\varphi_1, \xi, t'')\right) \quad (17)$$

$$\rho(\Diamond_I \varphi, \xi, t) = \sup_{t' \in t \ominus I} \rho(\varphi, \xi, t') \quad (18)$$

$$\rho(\Box_I \varphi, \xi, t) = \inf_{t' \in t \ominus I} \rho(\varphi, \xi, t'), \quad (19)$$

where $t \oplus I$ and $t \ominus I$ denote forward/backward time shifts. A trajectory satisfies φ iff $\rho(\varphi, \xi, t) > 0$, and $|\rho|$ measures the satisfaction or violation margin.

5.4 Smooth conjunctive robustness

The hard min in conjunctions has zero gradient with respect to all but the binding subformula, which is precisely the flat-objective pathology that defeated the prior hand-tuned ΔJ score: most candidate actions receive indistinguishable values. We therefore replace conjunction robustness by a smooth surrogate. With component robustness values $\{\rho_i\}_{i=1}^k$, the softmin

$$\tilde{\rho} = \text{softmin}_{\theta}(\rho_1, \dots, \rho_k) = -\frac{1}{\theta} \log \sum_{i=1}^k e^{-\theta \rho_i}, \quad \text{softmin}_{\theta} \xrightarrow{\theta \rightarrow \infty} \min, \quad (20)$$

is differentiable and assigns every component nonzero sensitivity, with $\theta > 0$ trading smoothness against fidelity to the true min. As an alternative we will also evaluate the arithmetic-geometric-mean (AGM) robustness of [5], which uses a geometric mean of $1 + \rho_i$ when all components are satisfied and the mean of the violated components otherwise, giving sound (under-approximating) yet gradient-bearing scores. We write $\tilde{\rho}(\Phi, \xi, [t, t + H])$ for the smooth robustness of specification Φ evaluated over the horizon window starting at t .

5.5 The four mission clauses

We weight and normalize each clause's predicate so the components are comparable in scale (dividing each margin by its threshold). The global specification is

$$\Phi = \varphi_{\text{exp}} \wedge \varphi_{\text{react}} \wedge \varphi_{\text{cov}} \wedge \varphi_{\text{hab}}. \quad (21)$$

(1) Bounded exposure φ_{exp} . Always, every monitored zone keeps its value-weighted exposure rate below a budget E^* :

$$\varphi_{\text{exp}} = \Box_{[0, T]} \bigwedge_Z (e_Z(t) \leq E^*), \quad f_{\text{exp}, Z}(t) = \frac{E^* - e_Z(t)}{E^*}. \quad (22)$$

(2) Reactive deadlines φ_{react} . Every reactive arrival is serviced within its deadline. Let $\text{arr}_a(t)$ pulse at arrival and $\text{srv}_a(t)$ be true once serviced:

$$\varphi_{\text{react}} = \Box_{[0,T]} \bigwedge_{a \in \mathcal{A}_{\text{react}}} (\text{arr}_a(t) \rightarrow \Diamond_{[0,T_{\text{react}}]} \text{srv}_a(t)). \quad (23)$$

Its predicate robustness is monotone in the deadline slack slack_a , so an action that services a reactive task improves φ_{react} in proportion to the slack recovered.

(3) Coverage recency φ_{cov} . No monitored cell goes unserved longer than T_{cov} :

$$\varphi_{\text{cov}} = \Box_{[0,T]} \bigwedge_Z (g_Z(t) \leq T_{\text{cov}}), \quad f_{\text{cov},Z}(t) = \frac{T_{\text{cov}} - g_Z(t)}{T_{\text{cov}}}. \quad (24)$$

(4) Anti-habitation / cue variety φ_{hab} . This is the history-dependent clause and is stated in two equivalent forms. The discrete past-time form forbids repeating a cue mode at a cell within a refractory window W :

$$\varphi_{\text{hab}}^{\text{pt}} = \Box_{[0,T]} \bigwedge_{Z,c} (\text{apply}_{Z,c}(t) \rightarrow \neg \Diamond_{(0,W]} \text{apply}_{Z,c}(t)), \quad (25)$$

where $\text{apply}_{Z,c}(t)$ pulses when a mode- c action completes in cell Z . The continuous effectiveness-floor form, preferred for robustness scoring because it is real-valued and smooth, requires that any active deterrence uses a sufficiently un-habituated mode:

$$\varphi_{\text{hab}} = \Box_{[0,T]} \bigwedge_Z (\text{active}_Z(t) \rightarrow \eta_{Z,c_Z^*(t)}(t) \geq \eta_{\min}), \quad f_{\text{hab},Z}(t) = \frac{\eta_{Z,c_Z^*(t)}(t) - \eta_{\min}}{1 - \eta_{\min}}, \quad (26)$$

where $c_Z^*(t)$ is the mode of the action currently applied in Z . A policy that repeats a cue drives $\eta_{Z,c}$ below $\eta_{\min}$ and incurs negative robustness; rotating modes or spacing applications in time keeps the margin positive. Equations (25) and (26) agree on the boolean semantics under the dynamics of (4)–(5); we use (26) for scoring and (25) for reporting variety compliance.

6 Robustness-Driven Predictive Task Value

6.1 Decentralized local specification

Each robot r monitors a local sub-specification Φ_r obtained by restricting the zone conjunctions in (21) to cells in $Z_r \cup \partial Z_r$; boundary signals are shared with neighbors over the existing EventBus. This keeps monitoring fully decentralized and matches the per-robot architecture: robot r needs only signals it observes or receives at its boundary.

6.2 Counterfactual robustness as task value

The value of a candidate predictive action $a = (x_a, c_a, r)$ is the improvement it makes to the local robustness over a planning horizon H . Let ξ^{-a} be the predicted local signal trajectory *without* a and ξ^{+a} the trajectory *with* a executed (completing near $t + \tau_{\text{service}}(a, r)$). The robustness-margin value is

$$U(a, r) = \tilde{\rho}(\Phi_r, \xi^{+a}, [t, t + H]) - \tilde{\rho}(\Phi_r, \xi^{-a}, [t, t + H]) \quad (27)$$

with an optional cost-normalized variant $U(a, r)/c_\eta(a, r)$. Equation (27) is the single quantity that replaces the prior ΔJ score everywhere in the pipeline.

Algorithm 1 Decentralized online robustness monitor (robot r , per tick)

Require: local cells $Z_r \cup \partial Z_r$, monitor period Δ_m , params $E^*, T_{\text{cov}}, T_{\text{react}}, \eta_{\min}$

- 1: **for** each cell $Z \in Z_r \cup \partial Z_r$ **do**
 - 2: update exposure e_Z from $\hat{\lambda}_r$ $\triangleright$ Eq. (8)
 - 3: recover effectiveness $\eta_{Z,c}$ for all c $\triangleright$ Eq. (4)
 - 4: age coverage $g_Z += \Delta_m$
 - 5: **end for**
 - 6: ingest boundary signals from neighbors over EventBus
 - 7: compute per-clause robustness margins (Eqs. (22),(23),(24),(26))
 - 8: **return** $\tilde{\rho}(\Phi_r, \xi, [t - \Delta_m, t])$ via smooth conjunction $\triangleright$ Eq. (20)
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6.3 Cheap analytic forward predictor

The counterfactual does not require a learned world model. Because ξ^{+a} and ξ^{-a} share the same assumed background and offspring dynamics over $[t, t + H]$, their difference is driven only by the suppression a injects and the effectiveness change it causes. Advancing the SESTPP field by passive decay over a lead δ ,

$$\hat{\lambda}_r(x, t + \delta) \approx [\mu(x)s(t + \delta) + \text{trigger}(x, t) e^{-\delta/\omega} - \text{inhib}(x, t) e^{-\delta/\omega_u}]_+, \quad (28)$$

the additional suppression contributed by a is

$$\Delta\lambda_a(x, t + \delta) = \min(\hat{\lambda}_r(x, t + \delta), \eta_{Z_a, c_a}(t) \beta_{c_a} K_{\sigma_u}(x - x_a) e^{-\delta/\omega_u}), \quad (29)$$

capped by the available intensity exactly as in the existing counterfactual estimator. This $\Delta\lambda_a$ feeds the three zone signals through closed forms: it lowers e_Z in (8) (raising φ_{exp} robustness), resets g_{Z_a} in (10) (raising φ_{cov} robustness), and (because executing mode c_a at Z_a applies the habituation jump (5)) moves η_{Z_a, c_a} in (26), *penalizing* a choice that would push effectiveness below $\eta_{\min}$. Thus a repeated, soon-to-habituate cue scores low even if it reduces instantaneous exposure, which is precisely the behavior a scalar exposure objective cannot produce.

6.4 Integration with the reserved-capacity dispatcher

The existing reserved-capacity dispatcher is retained unchanged except that its predictive utility is now $U(a, r)$ from (27). The reservation parameter ρ governs the average predictive bandwidth and a hard-latency override protects reactive deadlines; this proposal does not modify that logic, isolating the contribution to the value function.

7 Algorithms

8 Experimental Design

8.1 Testbed and protocol

All experiments run on the existing decentralized vineyard simulator with the established 24-hour headless sweeps and the seed set $\{123, 124, 125, 126, 127\}$, so results are directly comparable to the current per-run, per-metric, and advantage-vs-reference outputs. Reported quantities are paired across seeds with bootstrap confidence intervals, matching the existing summary tables.

Algorithm 2 Counterfactual predictive task value $U(a, r)$

Require: candidate $a = (x_a, c_a, r)$, horizon H , local spec Φ_r

- 1: predict nominal trajectory ξ^{-a} over $[t, t + H]$ ▷ Eq. (28)
 - 2: copy $\xi^{+a} \leftarrow \xi^{-a}$; inject $\Delta\lambda_a$ at completion lead ▷ Eq. (29)
 - 3: apply habituation jump to η_{Z_a, c_a} in ξ^{+a} ▷ Eq. (5)
 - 4: reset $g_{Z_a} \leftarrow 0$ in ξ^{+a} at completion
 - 5: $U \leftarrow \tilde{\rho}(\Phi_r, \xi^{+a}, [t, t + H]) - \tilde{\rho}(\Phi_r, \xi^{-a}, [t, t + H])$
 - 6: **return** U (or $U/c_\eta(a, r)$ for the cost-aware variant)
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Algorithm 3 Reserved-capacity dispatch with robustness value (robot r)

Require: queue Q_r , reservation ρ , window T_W , latency bound τ_{exp}

- 1: update sliding-window counters $t_{\text{react}}^r, t_{\text{pred}}^r, t_{\text{idle}}^r$; $\phi \leftarrow t_{\text{pred}}^r/T_W$
 - 2: $a_{\text{urg}} \leftarrow \arg \min_{a \in Q_r \cap \mathcal{A}_{\text{react}}} (t_a^{\text{exp}} - t)$
 - 3: **if** $a_{\text{urg}} \neq \emptyset$ **and** $t_{a_{\text{urg}}}^{\text{exp}} - t \leq \tau_{\text{exp}}$ **then**
 - 4: **return** a_{urg} ▷ hard-latency override
 - 5: **end if**
 - 6: **if** $\phi < \rho$ **and** $Q_r \cap \mathcal{A}_{\text{pred}} \neq \emptyset$ **then**
 - 7: **return** $\arg \max_{a \in Q_r \cap \mathcal{A}_{\text{pred}}} U(a, r)$ ▷ reservation claim, Eq. (27)
 - 8: **end if**
 - 9: **if** $Q_r \cap \mathcal{A}_{\text{react}} \neq \emptyset$ **then**
 - 10: **return** $\arg \max_{a \in Q_r \cap \mathcal{A}_{\text{react}}} v_a$ ▷ reactive service
 - 11: **end if**
 - 12: **if** $Q_r \cap \mathcal{A}_{\text{pred}} \neq \emptyset$ **then**
 - 13: **return** $\arg \max_{a \in Q_r \cap \mathcal{A}_{\text{pred}}} U(a, r)$ ▷ opportunistic
 - 14: **end if**
 - 15: **return** idle
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8.2 Baseline ladder

The ladder is designed to isolate each contribution.

ID	Predictive value	Purpose
B0	— (reactive only)	lower reference
B1	— (greedy unconstrained unc)	the policy the prior pipeline failed to beat
B2	SESTPP ΔJ	prior predictive value (the failed approach)
B3	STL robustness, $\Phi \setminus \varphi_{\text{hab}}$	isolates the robustness reformulation
B4	STL robustness, full Φ	full proposal (robustness + habituation)

B2–B4 all run under the identical reserved-capacity dispatcher with the same ρ , so any difference is attributable solely to the value function.

8.3 Factors

We sweep load regime $\rho_{\text{load}} \in \{0.5, 1.0, 1.5\}$ (spare, saturated, overloaded); habituation strength $\kappa \in \{\text{low}, \text{med}, \text{high}\}$ and recovery T_{rec} ; reservation $\rho \in \{0, 0.10, 0.25, 0.40\}$. The most important control is the *ground-truth habituation switch*: every condition is run once with the habituating truth process of (6) and once with habituation disabled ($\eta_{Z,c} \equiv 1$).

8.4 Metrics

Primary: value-weighted exposure J_{exp} (7); reactive deadline miss rate; cumulative coverage-violation time. **Specification:** empirical run-level robustness $\tilde{\rho}(\Phi)$ and each clause’s robustness, to show *which* requirement drives the outcome. **Habituation-specific:** mean effectiveness $\bar{\eta}$ at deterrence moments; a cue-variety index; and habituation-attributable exposure (the portion of J_{exp} traceable to low- η actions). **Efficiency:** travel distance, energy, tasks per km, communication overhead.

8.5 Hypotheses

- H1** Under habituating truth at $\rho_{\text{load}} \approx 1$, B4 reduces J_{exp} relative to B1 (predictive work finally pays).
- H2** $B4 > B3$ (the habituation clause adds value beyond the robustness reformulation alone).
- H3** $B3 \geq B2$ (smooth-robustness value, by avoiding the flat-gradient pathology, beats the hand-tuned ΔJ).
- H4** Under *non-habituating* truth, $B4 \approx B1$ (the advantage is specifically habituation-driven; this is the falsifiable control).

9 Implementation Plan

The work is localized; the SESTPP, zone partitioner, fleet model, and dispatcher core are reused. Estimated new/changed code is on the order of the prior reframe plus a monitor module.

Module	Change
SESTPP.py	add per-(cell,mode) effectiveness state and forward-decay helper for Eq. (28); effectiveness-scale the suppression injection
Robot.py	maintain local signal buffers ($e_Z, g_Z, \eta_{Z,c}$) and the recovery update (4)
planner_task_estimation.py	replace ΔJ estimator with counterfactual robustness $U(a, r)$ (Alg. 2)
planner_task_selection.py, planner_dispatch.py	swap predictive utility to $U(a, r)$; dispatcher logic unchanged
calibration_config.py	add monitor/habituation params $E^*, T_{\text{cov}}, T_{\text{react}}, W, \eta_{\text{min}}, T_{\text{rec}}, \{\kappa_c\}, \theta, H, \Delta_m$
DeterrentSystem.py (+ <code>_simple.tasks</code>)	extend the ground-truth suppression to Eq. (6) via the existing mode-dependent-suppression seam; wire monitor and metrics
telemetry_sim.py	log robustness (global and per-clause), $\bar{\eta}$, and variety metrics
new: stl_monitor.py	past/bounded-future STL robustness with smooth conjunction (20)
new: experiment driver	baseline ladder $\times$ factor grid, habituation on/off control

10 Scope, Timeline, and Threats to Validity

Scope. The thesis claims a habituation-aware spatio-temporal specification of the deterrence mission and a counterfactual-robustness predictive value function that, under realistic habituation

and near-saturated load, makes predictive multi-robot work pay where a scalar-exposure objective could not. It does *not* claim novelty of the point-process model, the zone partitioner, the simulator, or the reserved-capacity dispatcher; these are cited supporting infrastructure.

Indicative timeline. (1) Habituation model in truth process and forward predictor, plus the non-habituating control; (2) STL monitor module with smooth robustness and unit tests against reference robustness; (3) value-function integration and the B0–B4 ladder; (4) factor sweeps and analysis; (5) writing.

Risks and mitigations.

- *Flat task values* (the failure that motivated this work): mitigated by smooth robustness (20)/AGM and per-clause normalization.
- *Habituation too weak to be visible*: mitigated by scenario design with concentrated high-value zones and by sweeping κ ; the habituation-off control bounds the effect size.
- *Decentralized monitoring gaps at boundaries*: mitigated by sharing boundary signals over the existing EventBus and restricting Φ_r to $Z_r \cup \partial Z_r$.
- *Horizon/monitor-period sensitivity*: treated as secondary parameters, fixed at principled defaults and not swept in the headline experiments.

Threats to validity. The study is simulation-only with no physical-robot validation; the ground-truth process is known to the simulator and detection noise is not modeled; the predictive workload generator is in-house; and the habituation model is a stylized abstraction not fit to ecological field data. These limitations are framework-level and do not threaten the central claim, which is about the value function and is reproducible across alternative workload generators and habituation parameterizations.

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